

This is sample data provided for CALIBER purposes only		ONE POINT LESSON (OPL)		OPL No: OPL-CT-7801-04 Sheet 1 of 1
OPL Title	Circulating Water pH Control (AI-7805)	Discipline	Process / Operations	
Equipment	CT-7801 - COOLING TOWER CELL FAN	Area / Unit	7800 - COOLING TOWER SYSTEM	
Related Interlock	SEQ-7801 (SIL 1)	P&ID; Ref	TJC-LLD-PID-7801	
Classification:	☐ Basic Knowledge	☒ Improvement	☐ Trouble Case	Aspect: P-Q-C-D- S -M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **circulating water pH control (ai-7805)** for CT-7801 (COOLING TOWER CELL FAN) in the COOLING TOWER SYSTEM. Wrong pH causes scaling or corrosion across the entire cooling water loop, not just this cell. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **NON CRITICAL** handling **Circulating water / air** at up to atm / 50 degC. Observe the following before and during the task:

- Obtain the correct operating permit and brief the shift team before the task.
- Confirm hazardous fluids are isolated / inerted as required by procedure.
- Beware of flammable, hot, or high-pressure fluids; wear the correct PPE.
- Never defeat a safety interlock or trip without an authorised override permit.
- Have emergency actions and communications ready before starting.

3. TOOLS & MATERIALS REQUIRED

- Operating procedure & P&ID;
- Radio / communication with panel
- Gas detector (O2 / LEL) as required
- Correct PPE for the fluid handled
- Sample containers & PPE for sampling
- Permit-to-work documentation

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Read the AI-7805 pH each shift (target 7.0-8.5)	Permit valid & briefed
2	Check the acid / caustic dosing pumps are running	Parameter within limit
3	Cross-check against a manual lab titration	Interlock status confirmed
4	Adjust the dosing set-point as needed	Change logged on DCS
5	Report large pH swings to water treatment	Handover to next shift

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Routine barscreen cleaning on operator round	Normal debris accumulation	Raked barscreen clear, checked differential level
Scheduled fan gearbox oil change	Routine - oil sample acceptable	Drained and refilled ISO VG 220, replaced breather
VSHH-7802 vibration alarm on CT-7801 fan	Leading-edge erosion on one FRP blade caused imbalance	Inspected blades, re-balanced fan, vibration normalised

6. KEY LEARNING POINTS

- Wrong pH causes scaling or corrosion across the entire cooling water loop, not just this cell.
- Respect the interlock SEQ-7801: e.g. VSHH-7802 trips at > 9 mm/s - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-7801 before working on CT-7801.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
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